Bridge to the LQ Case

- Up to this point, the framework is completely general.
- The next question is:

  Can we solve this exploratory HJB explicitly?

- In general, the answer is difficult.
- But in the linear–quadratic setting:
  - the state dynamics are linear,
  - the reward is quadratic,
  - the HJB becomes tractable,
  - the optimal exploratory control turns out to be Gaussian.
- So the LQ case will give the first fully explicit answer to the continuous-time exploration problem.

The LQ State Dynamics: Linear in State and Control

Now, we assume the drift and volatility are linear in the state x and the control u:

$$b(x,u) = Ax + Bu, \qquad \sigma(x,u) = Cx + Du, \qquad x, u \in \mathbb{R}, \tag{4}$$

where

$$A, B, C, D \in \mathbb{R}.$$

- This is the continuous-time analogue of a linear controlled system.
- Interpretation:
  - A: state drift coefficient,
  - B: control effect in the drift,
  - C: state contribution to volatility,
  - D: control contribution to volatility.
- So both the deterministic part of the motion and the random/noise part of the motion, can depend on the control.

The LQ Reward: Quadratic in State and Control

The LQ reward function is assumed to be

$$r(x,u) = -\left(\frac{M}{2}x^2 + Rxu + \frac{N}{2}u^2 + Px + Qu\right), \qquad x,u \in \mathbb{R}, \tag{5}$$

where

$$M \geq 0, \qquad N > 0, \qquad R, P, Q \in \mathbb{R}.$$

- This is a quadratic reward (equivalently, a negative quadratic cost).
- Here, the signs are important:
  - $M \geq 0$: penalizes large state values,
  - $N > 0$: penalizes large controls,
  - R: couples state and control,
  - P, Q: linear terms.
- The strict condition $N > 0$ is especially important because it gives curvature in the control variable u.

Why the LQ Class Is So Important

- We emphasize that LQ control is central in classical control theory because it has:
  - elegant explicit solutions,
  - strong interpretability,
  - the ability to approximate more complex nonlinear problems.
- It is the canonical setting where one can:
  - solve the exploratory problem rigorously,
  - understand the role of entropy regularization,
  - identify the exact form of the optimal exploration distribution.
- This is also why LQ is the correct place to connect the theory with the common RL practice of using Gaussian exploration.

Why the Control Set Is Taken as $U = \mathbb{R}$

- We assume that the control set is **unconstrained**:

$$U = \mathbb{R},$$

  and this is standard in LQ control.
- Why is this natural?
  - quadratic optimization over $\mathbb{R}$ is analytically tractable,
  - the Gaussian distribution later obtained as optimal naturally lives on $\mathbb{R}$,
- Note:
  - if U were bounded, the HJB maximization would be more complicated,
  - but with $U = \mathbb{R}$, the exponential/Gaussian structure becomes transparent.

Definition 6.6

For each open-loop control distribution $\pi \in \mathcal{A}(x)$, we define its mean and variance processes by

$$\mu_t := \int_{\mathbb{R}} u\, \pi_t(u)\, du, \qquad \sigma_t^2 := \int_{\mathbb{R}} u^2\, \pi_t(u)\, du - \mu_t^2. \tag{6}$$

- So μ_t is the mean action at time t, and σ_t^2 is the action variance at time t.
- This is already a very important hint:
  - the mean will later encode **exploitation**,
  - the variance will later encode **exploration**.

Exploratory SDE in the LQ Model

- Starting from the general exploratory SDE

$$dX_t^\pi = \tilde{b}(X_t^\pi, \pi_t)\, dt + \tilde{\sigma}(X_t^\pi, \pi_t)\, dW_t,$$

and substituting the LQ coefficients $b(x, u) = Ax + Bu$ and $\sigma(x, u) = Cx + Du$, we obtain

$$dX_t^\pi = (AX_t^\pi + B\mu_t)\, dt + \sqrt{C^2(X_t^\pi)^2 + 2CDX_t^\pi \mu_t + D^2(\mu_t^2 + \sigma_t^2)}\, dW_t. \tag{7}$$

How Formula equation (7) Is Obtained

- We have seen that the exploratory coefficients are

$$\tilde{b}(y,\pi) = \int_{\mathbb{R}} b(y,u)\pi(u)\,du, \qquad \tilde{\sigma}^2(y,\pi) = \int_{\mathbb{R}} \sigma^2(y,u)\pi(u)\,du.$$

- Since $b(x,u) = Ax + Bu$, we get

$$\tilde{b}(x,\pi) = Ax + B\int_{\mathbb{R}} u\pi(u)\,du = Ax + B\mu_t.$$

- Since $\sigma(x,u) = Cx + Du$, we have

$$\sigma^2(x,u) = (Cx+Du)^2 = C^2x^2 + 2CDxu + D^2u^2.$$

- Averaging with respect to π_t,

$$\tilde{\sigma}^2(x,\pi_t) = C^2x^2 + 2CDx\mu_t + D^2\int_{\mathbb{R}} u^2\pi_t(u)\,du.$$

- Using

$$\int_{\mathbb{R}} u^2\pi_t(u)\,du = \mu_t^2 + \sigma_t^2,$$

we obtain formula (7).

More about Formula (7)

- The drift term is
$$AX_t^\pi + B\mu_t.$$
- So the state evolves according to the **mean action** μ_t, not according to one single deterministic control.
- The volatility term is
$$\sqrt{(CX_t^\pi + D\mu_t)^2 + D^2\sigma_t^2}.$$
- This shows that the diffusion depends on:
  - the current state X_t^π,
  - the mean control μ_t,
  - and the control variance σ_t^2.
- This is a very important conceptual point:
  In the exploratory LQ model, exploration affects not only the reward but also the volatility of the state process through the control variance.

The Entropy-Regularized Running Functional $L(X_t^\pi, \pi_t)$

- Define

$$L(X_t^\pi, \pi_t) := \int_{\mathbb{R}} r(X_t^\pi, u)\pi_t(u)\, du - \lambda \int_{\mathbb{R}} \pi_t(u) \ln \pi_t(u)\, du.$$

- This is the running reward appearing in the objective.
- It combines:
  - the averaged reward under the exploratory control,
  - the entropy regularization term.
- So this is the instantaneous quantity that balances:

$$\text{exploitation} \quad \text{vs.} \quad \text{exploration.}$$

- In the LQ setting, because $r(x, u)$ is quadratic in u, this quantity will later become explicit in terms of μ_t and σ_t^2.

The Case of State-Independent Reward

- We first consider the technically simpler case

$$r(x,u) = -\left(\frac{N}{2}u^2 + Qu\right), \qquad N > 0.$$

- Here the reward does **not** depend on the state x.
- So the immediate reward only measures the quality of the chosen action u, not the current location of the state process.
- In this case the state dynamics become **irrelevant** for the optimization itself.
- Why?
  - The objective depends only on the distribution of u,
  - so the current state does not directly affect the reward maximization step.
- Even so, the problem is still important in its own right, because it corresponds to the **state-independent RL problem**, also called the **continuous-armed bandit** problem in continuous time.

Why the State Dynamics Become Irrelevant Here

- Recall the entropy-regularized objective:

$$V(x) = \sup_{\pi \in \mathcal{A}(x)} \mathbb{E}\Big[\int_0^\infty e^{-\rho t}\Big(\int_{\mathbb{R}} r(X_t^\pi, u)\pi_t(u)\,du - \lambda \int_{\mathbb{R}} \pi_t(u) \ln \pi_t(u)\,du\Big)dt\Big]. \tag{8}$$

- Therefore,

$$r(X_t^\pi, u) = r(u) = -\left(\frac{N}{2}u^2 + Qu\right),$$

  so the reward term becomes

$$\int_{\mathbb{R}} r(u)\pi_t(u)\,du.$$

- Therefore, the state X_t^π no longer appears in the reward part of the objective.
- The only remaining optimization is over the **control distribution**.

The Feedback Control Distribution

- Recall that, the general feedback control distribution is

$$\pi^*(u;x) \propto \exp\left(\frac{1}{\lambda}\left[r(x,u) + \frac{1}{2}\sigma^2(x,u)v''(x) + b(x,u)v'(x)\right]\right).$$

- In the LQ model,

$$b(x,u) = Ax + Bu, \qquad \sigma(x,u) = Cx + Du,$$

  and in the state-independent reward case,

$$r(x,u) = -\left(\frac{N}{2}u^2 + Qu\right).$$

- Substituting these into formula (3) gives the candidate optimal feedback control.

Substituting the LQ Coefficients into Formula (3)

- Using

$$r(x,u) = -\left(\frac{N}{2}u^2 + Qu\right), \qquad b(x,u) = Ax + Bu,$$

and

$$\sigma(x,u) = Cx + Du,$$

the feedback control density becomes

$$\pi^*(u;x) = \frac{\exp\left(\frac{1}{\lambda}\left[-\left(\frac{N}{2}u^2 + Qu\right) + \frac{1}{2}(Cx+Du)^2 v''(x) + (Ax+Bu)v'(x)\right]\right)}{\int_{\mathbb{R}} \exp\left(\frac{1}{\lambda}\left[-\left(\frac{N}{2}\tilde{u}^2 + Q\tilde{u}\right) + \frac{1}{2}(Cx+D\tilde{u})^2 v''(x) + (Ax+B\tilde{u})v'(x)\right]\right) d\tilde{u}}. \tag{9}$$

- This is just formula (3) specialized to the current reward structure.

Expanding the Exponent in the Density

- Let us expand the u-dependent part of the exponent carefully.
- First,

$$\frac{1}{2}(Cx + Du)^2 v''(x) = \frac{1}{2}\left(C^2x^2 + 2CDxu + D^2u^2\right) v''(x).$$

- So the terms involving u are

$$-\frac{N}{2}u^2 - Qu + CDx\, u\, v''(x) + \frac{D^2}{2}u^2 v''(x) + Buv'(x).$$

- Grouping the u^2-terms and u-terms:

$$-\frac{1}{2}\big(N - D^2 v''(x)\big)u^2 + \big(CDx\, v''(x) + Bv'(x) - Q\big)u.$$

- Therefore the exponent is quadratic in u, which strongly suggests a Gaussian density.

Completing the Square in u

- Write
$$-\frac{1}{2}\big(N - D^2 v''(x)\big)u^2 + \big(CDx\, v''(x) + Bv'(x) - Q\big)u$$
in completed-square form.
- Set
$$a(x) := N - D^2 v''(x), \qquad b(x) := CDx\, v''(x) + Bv'(x) - Q.$$
- Then
$$-\frac{1}{2}a(x)u^2 + b(x)u = -\frac{a(x)}{2}\left(u - \frac{b(x)}{a(x)}\right)^2 + \frac{b(x)^2}{2a(x)}.$$
- The last term does not depend on u, so it is absorbed into the normalizing constant.
- Hence the density is proportional to
$$\exp\left(-\frac{1}{2\lambda}a(x)\left(u - \frac{b(x)}{a(x)}\right)^2\right).$$

The Candidate Optimal Distribution Is Gaussian

- Therefore the candidate optimal feedback control distribution is Gaussian:

$$\pi^*(u;x) = \mathcal{N}\left(u \,\middle|\, \frac{CDx\, v''(x) + Bv'(x) - Q}{N - D^2 v''(x)},\ \frac{\lambda}{N - D^2 v''(x)}\right). \tag{10}$$

- So at each current state x, the agent should explore according to a Gaussian distribution.
- We note that this calculation uses the condition

$$N - D^2 v''(x) > 0, \qquad x \in \mathbb{R},$$

which will be justified later.

Mean and Variance in the Gaussian Control Law

- The Gaussian density in (10) has:

$$\mu^*(x) = \frac{CDx\, v''(x) + Bv'(x) - Q}{N - D^2 v''(x)},$$

and

$$(\sigma^*(x))^2 = \frac{\lambda}{N - D^2 v''(x)}.$$

- Interpretation:
  - the mean $\mu^*(x)$ specifies the central action favored at state x,
  - the variance $(\sigma^*(x))^2$ specifies the exploration level around that action.
- Note that

$$\text{mean} \iff \text{exploitation}, \qquad \text{variance} \iff \text{exploration}.$$

Why the Positivity Condition Matters

- For a Gaussian density, the variance must be positive.
- Here the variance is
$$\frac{\lambda}{N - D^2 v''(x)}.$$
- Since
$$\lambda > 0,$$
we must have
$$N - D^2 v''(x) > 0.$$
- This condition is also needed so that the quadratic term in the exponent is negative:
$$-\frac{1}{2\lambda}\big(N - D^2 v''(x)\big)(u - \mu^*(x))^2.$$
- Otherwise the expression would not define an integrable density on $\mathbb{R}$.

Remark:state-independent reward case

- In the state-independent reward case, the exploratory HJB leads to a **Gaussian** candidate optimal control distribution.
- This is already a major theoretical result:
  - Gaussian exploration is not imposed by hand,
  - it emerges from solving the entropy-regularized optimization problem.
- The mean and variance are

$$\mu^*(x) = \frac{CDx\, v''(x) + Bv'(x) - Q}{N - D^2 v''(x)}, \qquad (\sigma^*(x))^2 = \frac{\lambda}{N - D^2 v''(x)}.$$

- The next step is to simplify these expressions further and identify the explicit value function and feedback law.

What Simplification Should We Expect Next?

- Recall that the reward is state-independent:

$$r(x, u) = -\left(\frac{N}{2}u^2 + Qu\right).$$

- Since the reward does not involve x, it is natural to expect that the value function may not need to depend on x either.
- If that happens, then

$$v'(x) = 0, \qquad v''(x) = 0.$$

- This would simplify the Gaussian control law dramatically.
- So the next step is to test whether a constant value function solves the HJB equation.

Trying a Constant Value Function

- Because the state does not appear in the reward then v is constant, and hence

$$v'(x) = 0, \qquad v''(x) = 0.$$

- Substituting these into the general Gaussian feedback law from the previous block gives

$$\mu^*(x) = \frac{-Q}{N}, \qquad (\sigma^*(x))^2 = \frac{\lambda}{N}.$$

- In particular, both the mean and the variance are independent of x.
- So the candidate optimal feedback control distribution becomes a **state-independent Gaussian distribution**.

The Candidate Optimal Distribution in Subsection 4.1

- The candidate optimal control distribution is therefore

$$\pi^*(u) = \mathcal{N}\left(u \,\middle|\, -\frac{Q}{N},\ \frac{\lambda}{N}\right).$$

- Interpretation:
  - the optimal exploratory mean action is

$$-\frac{Q}{N},$$

  - the optimal exploration variance is

$$\frac{\lambda}{N}.$$

- So in this state-independent problem, the optimal policy does not need to react to the state at all.

Computing the Optimal Value

- Since the optimal control distribution is now explicit, we can compute the optimal running reward directly.
- Under
$$\pi^*(u) = \mathcal{N}\left(-\frac{Q}{N}, \frac{\lambda}{N}\right),$$
we have
$$\mathbb{E}_{\pi^*}[u] = -\frac{Q}{N}, \qquad \mathrm{Var}_{\pi^*}(u) = \frac{\lambda}{N}.$$
- Therefore
$$\mathbb{E}_{\pi^*}[u^2] = \mathrm{Var}_{\pi^*}(u) + (\mathbb{E}_{\pi^*}[u])^2 = \frac{\lambda}{N} + \frac{Q^2}{N^2}.$$
- These are exactly the ingredients needed to compute the expected quadratic reward and the entropy term.

Expected Reward Under the Optimal Gaussian Law

- Recall
$$r(u) = -\left(\frac{N}{2}u^2 + Qu\right).$$
- Taking expectation under π^*,
$$\mathbb{E}_{\pi^*}[r(u)] = -\left(\frac{N}{2}\mathbb{E}_{\pi^*}[u^2] + Q\,\mathbb{E}_{\pi^*}[u]\right).$$
- Using
$$\mathbb{E}_{\pi^*}[u^2] = \frac{\lambda}{N} + \frac{Q^2}{N^2}, \qquad \mathbb{E}_{\pi^*}[u] = -\frac{Q}{N},$$
we obtain
$$\mathbb{E}_{\pi^*}[r(u)] = -\left(\frac{N}{2}\left(\frac{\lambda}{N} + \frac{Q^2}{N^2}\right) - \frac{Q^2}{N}\right).$$
- Simplifying,
$$\mathbb{E}_{\pi^*}[r(u)] = -\frac{\lambda}{2} + \frac{Q^2}{2N}.$$

Entropy of the Optimal Gaussian Law

- The entropy of a Gaussian density

$$\mathcal{N}(\mu, \sigma^2)$$

is

$$\mathcal{H} = \frac{1}{2} \ln(2\pi e \sigma^2).$$

- Here

$$\sigma^2 = \frac{\lambda}{N},$$

so

$$\mathcal{H}(\pi^*) = \frac{1}{2} \ln\left(2\pi e \frac{\lambda}{N}\right).$$

- Therefore the entropy contribution to the running objective is

$$\lambda \mathcal{H}(\pi^*) = \frac{\lambda}{2} \ln\left(2\pi e \frac{\lambda}{N}\right).$$

- This is the quantitative reward gained from maintaining exploration.

The Optimal Running Value and the Discounted Value Function

- The total optimal instantaneous objective is

$$\mathbb{E}_{\pi^*}[r(u)] + \lambda\mathcal{H}(\pi^*).$$

- From the previous slides,

$$\mathbb{E}_{\pi^*}[r(u)] = -\frac{\lambda}{2} + \frac{Q^2}{2N},$$

and

$$\lambda\mathcal{H}(\pi^*) = \frac{\lambda}{2}\ln\left(2\pi e\frac{\lambda}{N}\right).$$

- Hence the optimal running payoff is a constant, say c^*.
- Since the problem is infinite-horizon discounted, the value function is then

$$v(x) = \frac{c^*}{\rho},$$

which is again a constant, as expected.

The Case of State-Dependent Reward

- We now suppose the reward depends on both the state and the control:

$$r(x,u) = -\left(\frac{M}{2}x^2 + Rxu + \frac{N}{2}u^2 + Px + Qu\right), \qquad x, u \in \mathbb{R}.$$

- Compared with the state-independent reward case, two new terms now appear:
  - the pure state term $\frac{M}{2}x^2 + Px$,
  - the interaction term Rxu.
- Therefore:
  - the reward now distinguishes between different states,
  - the optimal exploratory control should in general depend on the current state x.
- This is the continuous-time analogue of the general RL problem, in contrast with the state-independent continuous-armed bandit case.

Assumption: A Condition on the Discount Rate

Assumption: The discount rate satisfies

$$\rho > 2A + C^2 + \max\left(\frac{D^2R^2 - 2NR(B + CD)}{N}, 0\right). \tag{11}$$

- In words, future rewards must be discounted strongly enough relative to the growth coming from:
  - the state drift coefficient A,
  - the state volatility coefficient C,
  - the state-control interaction through R and D.

Note on equation (11)

- If $R = 0$, then equation (11) reduces to the more familiar condition

$$\rho > 2A + C^2.$$

- This simpler condition already appears naturally when one wants discounted quadratic moments to remain controlled.
- More precisely, it helps ensure

$$\liminf_{T\to\infty} e^{-\rho T}\,\mathbb{E}\big[(X_T^\pi)^2\big] = 0,$$

  which is one of the admissibility conditions in the LQ problem.
- if ρ is too small, future growth of the state may dominate discounting,
- then the reward functional may fail to be finite or the verification argument may break down.
- So equation (11) is a **well-posedness/stability-type condition**.

Starting from the General Gaussian Formula

- Recall that the candidate optimal feedback control distribution has already been shown to be Gaussian in the LQ setting.
- Also, recall that the general formula for the state-independent case was

$$\pi^*(u;x) = \mathcal{N}\left(u \,\middle|\, \frac{CDx\, v''(x) + Bv'(x) - Q}{N - D^2 v''(x)},\ \frac{\lambda}{N - D^2 v''(x)}\right)$$

- In the present state-dependent case, the extra cross term

$$Rxu$$

changes the mean.
- So the next formula is the natural state-dependent analogue of the Gaussian law we already saw.

Candidate Optimal Feedback Control in dependent-case

The candidate optimal feedback control is

$$\pi^*(u;x) = \mathcal{N}\left(u \,\middle|\, \frac{CDx\, v''(x) + Bv'(x) - Rx - Q}{N - D^2 v''(x)},\ \frac{\lambda}{N - D^2 v''(x)}\right). \tag{12}$$

- The mean now contains the additional term

$$-Rx,$$

  coming from the state-control interaction in the reward.
- So the mean is now genuinely state-dependent.
- The variance still has the form

$$\frac{\lambda}{N - D^2 v''(x)}.$$

Mean and Variance from equation (12)

- Denote the mean and variance of $\pi^*(\cdot; x)$ by

$$\mu^*(x) = \frac{CDx\, v''(x) + Bv'(x) - Rx - Q}{N - D^2 v''(x)},$$

and

$$(\sigma^*(x))^2 = \frac{\lambda}{N - D^2 v''(x)}.$$

- Interpretation:
  - $\mu^*(x)$ tells us the central action selected at state x,
  - $(\sigma^*(x))^2$ tells us how much exploration is used around that action.
- Since $\mu^*(x)$ depends on x, the control law reacts to the current state.

Substituting π^* into the HJB

- We now substitute the Gaussian feedback control $\pi^*(\cdot; x)$ into the HJB equation (2).
- Denoting by $\mu^*(x)$ and $(\sigma^*(x))^2$ the mean and variance in (12), the HJB becomes

$$\rho v(x) = \int_{\mathbb{R}} r(x,u)\,\mathcal{N}\big(u \mid \mu^*(x), (\sigma^*(x))^2\big)\,du + \lambda \ln\Big(\sqrt{2\pi e\,(\sigma^*(x))^2}\Big)$$

$$+ v'(x) \int_{\mathbb{R}} (Ax + Bu)\,\mathcal{N}\big(u \mid \mu^*(x), (\sigma^*(x))^2\big)\,du$$

$$+ \frac{1}{2} v''(x) \int_{\mathbb{R}} (Cx + Du)^2\,\mathcal{N}\big(u \mid \mu^*(x), (\sigma^*(x))^2\big)\,du.$$

- This is the explicit Gaussian form of the exploratory HJB in the present subsection.

Why This HJB Form Is Useful

- Once the optimal control distribution is Gaussian, the integrals in the HJB become expectations under a Gaussian law.
- Therefore they can be evaluated using only:
  - the Gaussian mean $\mu^*(x)$,
  - the Gaussian variance $(\sigma^*(x))^2$.
- In particular, we only need the standard Gaussian moment formulas:

$$\mathbb{E}[U] = \mu^*(x), \qquad \mathbb{E}[U^2] = (\sigma^*(x))^2 + (\mu^*(x))^2.$$

- This is why the LQ structure is so tractable:
  - the reward is quadratic in u,
  - the dynamics are linear in u,
  - Gaussian moments close the equation.

Evaluating the Gaussian Expectations in the HJB

- Let
$$U \sim \mathcal{N}\big(\mu^*(x), (\sigma^*(x))^2\big).$$
Then
$$\mathbb{E}[U] = \mu^*(x), \qquad \mathbb{E}[U^2] = (\mu^*(x))^2 + (\sigma^*(x))^2.$$
- Therefore:
$$\int_{\mathbb{R}} (Ax + Bu)\,\mathcal{N}(du) = Ax + B\mu^*(x),$$
and
$$\int_{\mathbb{R}} (Cx + Du)^2\,\mathcal{N}(du) = (Cx + D\mu^*(x))^2 + D^2(\sigma^*(x))^2.$$
- Similarly, because the reward is quadratic in u, its Gaussian expectation can also be written explicitly in terms of $\mu^*(x)$ and $(\sigma^*(x))^2$.

The Reduced HJB Equation

- After computing the Gaussian expectations and reorganizing the terms, we obtain

$$\begin{aligned}\rho v(x) = &\frac{\big(CDx\, v''(x) + Bv'(x) - Rx - Q\big)^2}{2\big(N - D^2 v''(x)\big)} \\ &+ \frac{\lambda}{2}\left(\ln\left(\frac{2\pi e\lambda}{N - D^2 v''(x)}\right) - 1\right) \\ &+ \frac{1}{2}\big(C^2 v''(x) - M\big)x^2\big(Av'(x) - P\big)x.\end{aligned} \tag{13}$$

- This is the reduced HJB equation for the state-dependent LQ problem.
- All dependence on the control distribution has now disappeared.
- The unknown is only the value function v.

Quadratic guess for the Value Function

- Under (11) and the additional condition

$$R^2 < MN,$$

one smooth solution to the HJB equation (13) is given by

$$v(x) = \frac{1}{2}k_2x^2 + k_1x + k_0.$$

- Then

$$v'(x) = k_2x + k_1, \qquad v''(x) = k_2.$$

- So the second derivative is constant, which greatly simplifies (13).

A More General Quadratic guess

- In general, one may start from the generic quadratic guess

$$v(x) = \frac{1}{2}k_2x^2 + k_1x + k_0, \qquad x \in \mathbb{R}.$$

- Substituting this into (13) yields a system of algebraic equations for (a_2, a_1, a_0).
- One can see these equations explicitly as

$$\rho k_2 = \frac{(k_2(B + CD) - R)^2}{N - k_2D^2} + k_2(2A + C^2) - M,$$

$$\rho k_1 = \frac{(k_1B - Q)(k_2(B + CD) - R)}{N - k_2D^2} + k_1A - P,$$

$$\rho k_0 = \frac{(k_1B - Q)^2}{2(N - k_2D^2)} + \frac{\lambda}{2}\left(\ln\left(\frac{2\pi e\lambda}{N - k_2D^2}\right) - 1\right).$$

- So the PDE has been reduced to a finite-dimensional algebraic system.

The Explicit Coefficient k_2

- The explicit solution coefficient:

$$\Delta := \big((\rho - (2A + C^2))N + 2(B + CD)R - D^2 M\big)^2 - 4\big((B + CD)^2 + (\rho - (2A + C^2))D^2\big)(R^2 - MN),$$

and

$$D_0 := (B + CD)^2 + (\rho - (2A + C^2))D^2.$$

Then

$$k_2 = \frac{(\rho - (2A + C^2))N + 2(B + CD)R - D^2 M - \sqrt{\Delta}}{2D_0}. \tag{14}$$

The Coefficients k_1 and k_0

- Once k_2 is known, the other coefficients are

$$k_1 := \frac{P(N - k_2 D^2) - QR}{k_2 B(B + CD) + (A - \rho)(N - k_2 D^2) - BR}, \tag{15}$$

and

$$k_0 := \frac{(k_1 B - Q)^2}{2\rho(N - k_2 D^2)} + \frac{\lambda}{2\rho}\left(\ln\left(\frac{2\pi e \lambda}{N - k_2 D^2}\right) - 1\right). \tag{16}$$

- So the candidate value function is fully explicit:

$$v(x) = \frac{1}{2} k_2 x^2 + k_1 x + k_0.$$

A Key Property: $k_2 < 0$

- For this particular solution,

$$k_2 < 0,$$

due to (13) and the condition $R^2 < MN$.

- Therefore the value function

$$v(x) = \frac{1}{2}k_2 x^2 + k_1 x + k_0$$

is **concave** in x.

- Also,

$$N - D^2 v''(x) = N - k_2 D^2 > 0,$$

so the Gaussian variance is well-defined.

- This is exactly the positivity condition we needed earlier for the candidate optimal Gaussian law.

Theorem

Theorem 6.1

Suppose

$$r(x,u) = -\left(\frac{M}{2}x^2 + Rxu + \frac{N}{2}u^2 + Px + Qu\right),$$

with

$$M \geq 0, \qquad N > 0, \qquad R, Q, P \in \mathbb{R}, \qquad R^2 < MN,$$

and suppose (13) holds. Then the value function in (8) is

$$V(x) = \frac{1}{2}k_2x^2 + k_1x + k_0, \qquad x \in \mathbb{R},$$

where k_2, k_1, k_0 are given by (14), (15), and (16), respectively. Moreover, the optimal feedback control is Gaussian with density

$$\pi^*(u;x) = \mathcal{N}\left(u \,\middle|\, \frac{(k_2(B+CD)-R)x + k_1B - Q}{N - k_2D^2}, \frac{\lambda}{N - k_2D^2}\right).$$

Remark

- Theorem 6.1 says that in the general state-dependent LQ problem:
  - the value function is quadratic in the state,
  - the optimal feedback control distribution is Gaussian.
- The Gaussian mean is

$$\mu^*(x) = \frac{(k_2(B + CD) - R)x + k_1 B - Q}{N - k_2 D^2},$$

  which depends on the current state x.
- The Gaussian variance is

$$(\sigma^*)^2 = \frac{\lambda}{N - k_2 D^2},$$

  which is **state-independent**.
- This is already a striking structural result:

  state affects the mean, but not the variance.

The Associated Optimal State Process

- Finally, the optimal state process X_t^* under $\pi^*(\cdot;\cdot)$ is the unique solution of

$$dX_t^* = \left[\left(A + \frac{B(k_2(B+CD)-R)}{N-k_2D^2}\right)X_t^* + \frac{B(k_1B-Q)}{N-k_2D^2}\right]dt$$
$$+\sqrt{\left[\left(C + \frac{D(k_2(B+CD)-R)}{N-k_2D^2}\right)X_t^* + \frac{D(k_1B-Q)}{N-k_2D^2}\right]^2 + \frac{\lambda D^2}{N-k_2D^2}}\,dW_t, \tag{17}$$

- This is the optimal exploratory state dynamics corresponding to the Gaussian feedback control. :contentReference[oaicite:9]index=9

The Optimal Feedback Control Rewritten

- From Theorem 6.1, the optimal feedback control is

$$\pi^*(u;x) = \mathcal{N}\left(u \;\middle|\; \frac{(k_2(B+CD)-R)x + k_1 B - Q}{N - k_2 D^2}, \frac{\lambda}{N - k_2 D^2}\right).$$

- So we can write

$$\pi^*(u;x) = \mathcal{N}\big(u \mid \mu^*(x), (\sigma^*)^2\big),$$

where

$$\mu^*(x) = \frac{(k_2(B+CD)-R)x + k_1 B - Q}{N - k_2 D^2},$$

and

$$(\sigma^*)^2 = \frac{\lambda}{N - k_2 D^2}.$$

- This notation makes the interpretation much clearer.